

**1. The delivery times of five online orders are: 20, 25, 30, 35, 40 minutes
Calculate the mean and population variance.**

Given delivery times: 20, 25, 30, 35, 40 minutes

Step 1: Calculate the Mean

Formula:

Mean = Sum of observations / Number of observations

$$\text{Mean} = (20 + 25 + 30 + 35 + 40) / 5$$

$$\text{Mean} = 150 / 5 = 30$$

Mean = 30 minutes

Step 2: Calculate the Population Variance

Formula:

$$\text{Population Variance} = \Sigma(X - \mu)^2 / N$$

Where:

X = Each observation

μ = Population mean

N = Number of observations

Delivery Time (X)	X - 30	(X - 30)²
20	-10	100
25	-5	25
30	0	0
35	5	25
40	10	100

Sum of squared deviations:

$$100 + 25 + 0 + 25 + 100 = 250$$

Population Variance:

$$\text{Variance} = 250 / 5 = 50$$

Population Variance = 50 minutes²

2. A university has 5,000 students. A random sample of 100 students is selected to estimate the average daily study time. Explain why sampling is used instead of surveying all 5,000 students.

Given:

Total number of students = 5,000

Sample size = 100 students

A university could survey all 5,000 students, but this would require more time, money, effort, and resources.

Instead, a random sample of 100 students is selected. If the sample is properly selected, it can provide a good estimate of the average daily study time of all 5,000 students.

Reasons for Using Sampling:

1. Saves time: Collecting data from 100 students is faster than collecting data from 5,000 students.
2. Reduces cost: Fewer students require fewer resources and less money.
3. Easier to manage: A smaller sample is easier to collect, organize, and analyze.
4. Provides useful estimates: A properly selected random sample can represent the larger population.
5. Practical: Surveying every student may be difficult or unnecessary.

Conclusion:

Sampling is used because it is faster, cheaper, easier, and more practical than surveying the entire population of 5,000 students, while still providing a reliable estimate when the sample is properly selected.

3. From a sample of 200 customers, 150 customers are satisfied with a product. Calculate the sample proportion of satisfied customers and express it as a percentage.

Given:

Total sample = 200 customers

Satisfied customers = 150

Formula:

Sample Proportion = Number of satisfied customers / Total number of customers

$$p = 150 / 200$$

$$p = 0.75$$

Sample Proportion = 0.75

Convert to Percentage:

$$\text{Percentage} = 0.75 \times 100$$

$$\text{Percentage} = 75\%$$

Percentage of satisfied customers = 75%

Therefore, 75% of the sampled customers are satisfied with the product.

4. Calculate the covariance between the following

X	Y
2	5
4	7
6	9
8	11

Determine whether the relationship is positive or negative.

Step 1: Calculate the Mean of X

Formula:

$$\text{Mean of X} = \Sigma X / N$$

$$= (2 + 4 + 6 + 8) / 4$$

$$= 20 / 4 = 5$$

Step 2: Calculate the Mean of Y

$$\begin{aligned}\text{Mean of Y} &= \Sigma Y / N \\ &= (5 + 7 + 9 + 11) / 4 \\ &= 32 / 4 \\ &= 8\end{aligned}$$

Step 3: Calculate Covariance

For population covariance:

$$\text{Cov}(X, Y) = \Sigma[(X - \bar{X})(Y - \bar{Y})] / N$$

X	Y	X - 5	Y - 8	(X - 5) (Y - 8)
2	5	-3	-3	9
4	7	-1	-1	1
6	9	1	1	1
8	11	3	3	9

Sum of products:

$$9 + 1 + 1 + 9 = 20$$

Covariance:

$$\text{Cov}(X, Y) = 20 / 4 = 5$$

Answer:

Covariance = 5

Since the covariance is positive, X and Y have a positive relationship.

As X increases, Y also increases.

Conclusion: Positive relationship.

5. The following data represents hours of exercise (X) and calories burned (Y).

Exercise Hours (X)	Calories Burned (Y)
1	200
2	300
3	400
4	500
5	600

Calculate the covariance and interpret the result.

Step 1: Calculate Mean of X

$$\begin{aligned}
 \text{Mean of X} &= (1 + 2 + 3 + 4 + 5) / 5 \\
 &= 15 / 5 \\
 &= 3
 \end{aligned}$$

Step 2: Calculate Mean of Y

$$\begin{aligned}
 \text{Mean of Y} &= (200 + 300 + 400 + 500 + 600) / 5 \\
 &= 2000 / 5 \\
 &= 400
 \end{aligned}$$

Step 3: Calculate Covariance

Formula:

$$\text{Cov}(X,Y) = \Sigma[(X - \bar{X})(Y - \bar{Y})] / N$$

X	Y	X - 3	Y - 400	(X - 3)(Y - 400)
1	200	-2	-200	400
2	300	-1	-100	100
3	400	0	0	0
4	500	1	100	100
5	600	2	200	400

Sum of products: $400 + 100 + 0 + 100 + 400 = 1000$

Covariance:

$$\text{Cov}(X,Y) = 1000 / 5 = 200$$

Answer:

Covariance = 200

Since the covariance is positive, there is a positive relationship between exercise hours and calories burned.

As exercise hours increase, calories burned also increase.

Conclusion: Exercise hours and calories burned have a positive relationship.

6. Calculate Pearson's correlation coefficient for the following dataset and interpret the strength and direction of the relationship.

Study Hours (X)	Marks (Y)
1	45
2	50
3	55
4	65
5	70

Formula

Pearson's correlation coefficient:

$$r = \Sigma[(X - \bar{X})(Y - \bar{Y})] / \sqrt{[\Sigma(X - \bar{X})^2 \times \Sigma(Y - \bar{Y})^2]}$$

Step 1: Calculate Mean of X

$$\begin{aligned}\bar{X} &= (1 + 2 + 3 + 4 + 5) / 5 \\ &= 15 / 5 \\ &= 3\end{aligned}$$

Step 2: Calculate Mean of Y

$$\begin{aligned}\bar{Y} &= (45 + 50 + 55 + 65 + 70) / 5 \\ &= 285 / 5 = 57\end{aligned}$$

Step 3: Prepare the Table

X	Y	X - 3	Y - 57	(X - 3) ²	(Y - 57) ²	(X - 3)(Y - 57)
1	45	-2	-12	4	144	24
2	50	-1	-7	1	49	7
3	55	0	-2	0	4	0
4	65	1	8	1	64	8
5	70	2	13	4	169	26

Therefore:

$$\Sigma(X - \bar{X})^2 = 10$$

$$\Sigma(Y - \bar{Y})^2 = 430$$

$$\Sigma(X - \bar{X})(Y - \bar{Y}) = 65$$

Step 4: Calculate Pearson's r

$$r = 65 / \sqrt{(10 \times 430)}$$

$$= 65 / \sqrt{4300}$$

$$= 65 / 65.574$$

$$r \approx 0.991$$

Pearson's correlation coefficient = 0.991

Interpretation

The value of r ranges from -1 to +1.

Here:

$$r = 0.991$$

Since r is very close to +1, there is a very strong positive correlation between study hours and marks.

This means that as study hours increase, marks also tend to increase.

Final Answer: $r \approx 0.991$

Strength: Very strong

Direction: Positive

Conclusion: Students who study more hours tend to obtain higher marks.